Probabilistic Shaping for Clipping- and Peak-Limited LED Links: Recent Progress

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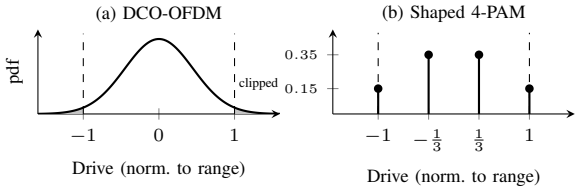


Fig. 1. The two constraint models (illustration drawn for this survey; dashed: LED range). (a) The near-Gaussian DCO-OFDM signal is clipped. (b) A PAM constellation fits the range, so shaping acts on the peak.

I. Introduction

Visible light communication (VLC) and short-reach optical interconnects use intensity modulation with direct detection (IM/DD): the signal must be real and non-negative, and the LED or modulator is linear only over a limited current range. Multicarrier formats handle the frequency-selective LED response well, but their high peak-to-average power ratio (PAPR) forces clipping. Recent work adapts probabilistic constellation shaping (PCS) to such clipping- and peak-limited links. The chosen base paper [1] optimizes PCS for clipped DC-biased optical OFDM (DCO-OFDM). This survey reviews 26 journal papers from 2021–2026; Fig. 1 illustrates the problem they address.

II. Optical OFDM and the Clipping Problem

Zhang *et al.* [2] survey the evolution of optical OFDM for both VLC and fiber, and position layered ACO-OFDM between ACO- and DCO-OFDM in the trade-off between spectral and power efficiency. Bai and Hranilovic [3] propose a layered antisymmetry-constructed clipped OFDM that needs less than half of the multiplications and additions of layered ACO-OFDM. Pham and Ishihara [1] show that PCS can increase the PAPR of DCO-OFDM and thus the clipping distortion; they formulate capacity maximization with clipping, which is non-convex, and solve it sub-optimally by projected gradient descent with a single shaping distribution for all subcarriers. Alrakah *et al.* [4] analyse clipping noise in OFDM-based VLC and show that DCO-OFDM with PAPR reduction achieves a lower BER than conventional DCO-OFDM.

An alternative is to avoid the Gaussian-like OFDM waveform. Hussin and Shalaby [5] compare DFT-spread OFDM with OFDM for an indoor single-LED link in terms of BER and PAPR. Saied, Li and Rabie [6] propose DFT-spread optical PAM, which in simulation has about 7 dB lower PAPR than DCO-OFDM and in experiment needs about 2.5 dB less average transmit power. For vehicular VLC, Saied *et al.* [7] report a 9 dB PAPR advantage of DFT-spread PAM, but find that DCO-OFDM is better when the signal bandwidth greatly exceeds the LED modulation bandwidth.

III. Shaping Under Peak Constraints

Without optical amplification, an IM/DD link is limited by peak rather than average power. Wiegart *et al.* [8] optimize the input distribution of 4-PAM under the peak constraint set by the modulator's extinction ratio, and confirm gains in reach and rate adaptability experimentally. Che, Cho and Chen [9] show that a peak-constrained system can be partly or fully converted into an average-constrained one, where conventional PCS helps, if the transmitter DSP raises the PAPR; they verify this with 4- and 8-PAM. Che and Chen find that faster-than-Nyquist PAM-4 gains over higher-order formats on gradually decaying channels [10], and review IM-DD modulation and DSP beyond 200 Gb/s [11]. Liang and Kahn [12] use a high-rate continuous approximation to obtain closed-form capacity-achieving inputs and shaping gains for optical channels, including direct detection; Liang *et al.* [13] add peak and average power constraints and note that PS must expand the peak-to-average ratio, which complicates peak-limited systems.

In VLC, Gutema *et al.* [14] demonstrate a three-LED WDM link with probabilistically shaped OFDM at 10.81 Gb/s, 25 % above adaptive bit-and-power loading. Nguyen *et al.* [15] jointly optimize the PAM symbol probabilities and the precoder of a multi-user VLC broadcast channel under amplitude constraints, with sum-rate gains of about 17.5 % (8-PAM) and 19.2 % (16-PAM) over uniform signaling. Pham and Ishihara [16] use PCS of M-PAM for physical-layer security in VLC and obtain a design that meets the BER requirement over the whole optical power range, where uniform signaling does not. Gao *et al.* [17] design multidimensional geometrically shaped constellations for MISO VLC under per-LED optical power constraints at finite blocklength.

IV. Practical Shaping and Distribution Matching

Short distribution matchers make PCS implementable. Chen *et al.* [18] optimize enumerative sphere shaping, beating

TABLE I
REPRESENTATIVE WORKS AND THEIR CONSTRAINT MODEL

Ref.	Waveform	Constraint	Main contribution
[1]	DCO-OFDM	Clipping	PCS via PGD
[6]	DFT-s PAM	LED range	7 dB lower PAPR
[4]	DCO-OFDM	Clipping	PAPR reduction, BER
[15]	PAM	Amplitude	PCS + precoding
[13]	DD, PAM	Peak + avg.	Shaping gains
[17]	MISO VLC	Per-LED power	Multidimensional GS

constant-composition matching at short blocklengths. Matsumine *et al.* [19] realize probabilistic amplitude shaping with short linear block codes and report 0.3–1.0 dB shaping gain for blocks shorter than 24 symbols. For multicarrier links, Liu *et al.* [20] flatten the uneven subcarrier SNR with partial Walsh–Hadamard precoding, cutting the real multiplications from $2P^2 - 2P$ to $2P \log_2 P$ with about 1 dB sensitivity gain over conventional PS. Autoencoder-learned constellations are more robust to channel uncertainty than QAM [21]. In OFDM sensing and communication (ISAC), Geiger *et al.* [22] compare geometric, probabilistic and joint shaping and Du *et al.* [23] extend PCS to mismatched-filter sensing; Guo *et al.* [24] reach 101.47 b/s/Hz with PS-16QAM over 7-core 3-mode fiber.

V. LED NONLINEARITY

Inside its range, the LED is also nonlinear. Deng *et al.* [25] build a physics-based LED model and a low-complexity post-compensator implemented in real time on an FPGA, and Park *et al.* [26] compensate LED nonlinearity in DCO-OFDM VLC with coefficient-approximation predistortion and a bidirectional LSTM.

VI. OPEN QUESTIONS AND CONCLUSION

Three trends emerge: shaping is moving to peak- and amplitude-constrained models [8], [9], [13], [15], [17]; DFT-spread waveforms are reconsidered for LEDs for their lower PAPR [6], [7]; and short matchers and learned shaping make PCS practical [18], [19], [21]. Two questions remain open in the reviewed papers. First, OFDM shaping under clipping [1] uses one distribution for all subcarriers and does not load shapes and powers jointly with the clipping operating point. Second, peak-constrained PAM shaping has not been compared directly with an optimized clipped DCO-OFDM link of the same bandwidth and LED range.

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